GSSS INSTITUTE OF ENGINEERING & TECHNOLOGY FOR WOMEN, MYSURU
Department of Electronics & Communication Engineering | GATE Preparation Library

Communication Systems — Official GATE ECE Syllabus

Paper Code: EC • Section 7: Communications (9 - 12 Marks)

Weightage: 9 - 12 Marks • **Subject Code:** COMM • **Key Focus:** Random Variables, Analog/Digital Modulation, Noise, Information Theory

OFFICIAL GATE EXAMINATION PATTERN & MARKING SCHEME
• **Question Types:** Multiple Choice Questions (MCQ, 1/3 negative marks), Multiple Select Questions (MSQ, zero negative marks), Numerical Answer Type (NAT, zero negative marks).
• **Weightage:** General Aptitude (15 Marks) + Engineering Mathematics (13 Marks) + Core Technical Subjects (72 Marks) = **100 Total Marks (65 Questions, 180 Minutes)**.

SECTION: RANDOM SIGNALS, PROCESSES & NOISE

- Random processes: Continuous-time random process, ensemble averages, stationarity: Wide-Sense Stationary (WSS) and Strict-Sense Stationary (SSS).
- Autocorrelation function (ACF) R_xx(tau): Properties (R_xx(0) = total average power, even symmetry R_xx(tau) = R_xx(-tau), maximum at origin |R_xx(tau)| <= R_xx(0)).
- Cross-correlation function and cross-spectral density.
- Ergodicity: Mean-ergodic and correlation-ergodic processes (time averages equal ensemble averages).
- Power Spectral Density (PSD) S_xx(f): Wiener-Khinchin Theorem (PSD is the Fourier transform of the autocorrelation function), properties of PSD (S_xx(f) >= 0, even function for real processes).
- Transmission of random signals through linear time-invariant (LTI) systems: S_yy(f) = S_xx(f) * |H(f)|^2, R_yy(tau) = R_xx(tau) * h(tau) * h(-tau).
- Noise characterization: Thermal noise (Johnson noise, PSD = 2*k*T*R), White Gaussian Noise (AWGN, flat two-sided PSD = N0/2, zero mean, uncorrelated samples).
- Narrowband noise representation: In-phase and quadrature components n(t) = n_I(t)*cos(2*pi*fc*t) - n_Q(t)*sin(2*pi*fc*t). Statistical properties of n_I(t) and n_Q(t).
- Noise figure (F), equivalent noise temperature (Te = T0*(F - 1)), Friis formula for cascaded stages: F_total = F1 + (F2 - 1)/G1 + (F3 - 1)/(G1*G2).

SECTION: ANALOG COMMUNICATIONS (MODULATION & DEMODULATION)

- Amplitude Modulation (AM): Standard AM (DSB-FC) time and frequency domain expression, modulation index (mu = Am / Ac).
- Total transmitted power: Pt = Pc * (1 + mu^2 / 2), transmission efficiency (eta = mu^2 / (2 + mu^2), max efficiency = 33.3% for single tone sinusoidal modulation).
- Double Sideband Suppressed Carrier (DSB-SC): Time and frequency domain expressions, 100% power efficiency, bandwidth = 2 * fm. Coherent demodulation and Costas loop.
- Single Sideband Suppressed Carrier (SSB-SC): Hilbert transform concept (90 degree phase shift, frequency response H(f) = -j*sgn(f)), time domain expression s(t) = Ac*m(t)*cos(wc*t) +- Ac*m_hat(t)*sin(wc*t). Bandwidth = fm (most bandwidth efficient analog scheme).
- Vestigial Sideband Modulation (VSB): Sideband shaping filter transfer function condition: H(f + fc) + H(f - fc) = constant for |f| <= fm. Application in television video broadcast.
- Demodulation of AM: Envelope detector (RC time constant constraint for preventing diagonal clipping: RC <= sqrt(1 - mu^2) / (mu * wm)), square-law detector.
- Angle Modulation: Frequency Modulation (FM) and Phase Modulation (PM). Instantaneous phase and frequency definitions. Frequency deviation (delta_f = kf * Am), modulation index (beta = delta_f / fm).
- Narrowband FM (beta << 1, bandwidth = 2*fm) vs. Wideband FM (Bessel function expansion of frequency spectrum).
- Carson's Rule for transmission bandwidth: BW = 2 * (delta_f + fm) = 2 * fm * (beta + 1).

• Generation of FM: Direct method (varactor diode reactance modulator) and Indirect method (Armstrong method using frequency multipliers).

• Demodulation of FM: Slope detector, balanced slope detector, Foster-Seeley discriminator, Phase-Locked Loop (PLL) FM demodulator.

• Superheterodyne receiver: RF amplifier, mixer, local oscillator (LO), Intermediate Frequency (IF = 455 kHz for AM, 10.7 MHz for FM). Image frequency: f_img = f_signal + 2*IF. Image Rejection Ratio (IRR).

• Noise performance of analog systems: Figure of Merit (FOM = SNR_output / SNR_channel). FOM for DSB-SC = 1, SSB-SC = 1, AM = mu^2 / (2 + mu^2).

• Threshold effect in envelope detection and FM discriminator. Pre-emphasis and De-emphasis circuits for boosting SNR in high-frequency audio FM.

SECTION: INFORMATION THEORY & SOURCE CODING

• Measure of information: Self-information I(xi) = log2(1 / P(xi)) bits.

• Entropy of a discrete memoryless source: H(X) = -sum(P(xi) * log2 P(xi)) in bits/symbol. Maximum entropy for M equiprobable symbols: H_max = log2(M).

• Entropy of binary source: H(p) = -p*log2(p) - (1-p)*log2(1-p).

• Source coding theorem (Shannon's first theorem): Average codeword length L >= H(X). Coding efficiency eta = H(X) / L.

• Source coding algorithms: Shannon-Fano coding, Huffman coding (optimal prefix-free variable length code construction).

• Mutual information: I(X; Y) = H(X) - H(X|Y) = H(Y) - H(Y|X). Properties: I(X; Y) >= 0, symmetric.

• Channel Capacity (Shannon's second theorem): C = max I(X; Y). Discrete Memoryless Channels: Binary Symmetric Channel (BSC, capacity C = 1 - H(p)), Binary Erasure Channel (BEC, capacity C = 1 - alpha).

• Shannon-Hartley Theorem for AWGN channel: C = B * log2(1 + S / (N0 * B)) in bits/second. Capacity in the limit of infinite bandwidth: C_infinity = (S / N0) * log2(e) = 1.442 * (S / N0).

SECTION: DIGITAL BASEBAND TRANSMISSION & PULSE SHAPING

• Pulse Code Modulation (PCM) transmitter and receiver: Sampling, Uniform Quantization, Encoding.

• Quantization step size: delta = (V_max - V_min) / L where L = 2^n is the number of quantization levels.

• Quantization noise power: Nq = delta^2 / 12 for uniform quantizer.

• Signal to Quantization Noise Ratio for sinusoidal input: (SNR)_dB = 6.02 * n + 1.76 dB (each additional bit increases SNR by ~6 dB).

• Non-uniform quantization: Companding (Compressor + Uniform Quantizer + Expander). mu-law companding (North America/Japan) and A-law companding (Europe/India).

• Differential Pulse Code Modulation (DPCM): Prediction filter, transmits prediction error rather than raw sample, reduces transmission bandwidth.

• Delta Modulation (DM): 1-bit quantizer (step size delta). Slope Overload Distortion (occurs when slope of signal exceeds maximum tracking rate: |dm(t)/dt| > delta * fs), Granular Noise (occurs when step size is too large for slowly varying signals). Adaptive Delta Modulation (ADM) solution.

• Line coding techniques: Unipolar NRZ/RZ, Polar NRZ/RZ, Bipolar NRZ (AMI - Alternate Mark Inversion), Manchester coding (biphase, self-clocking, zero DC component). Power spectra comparison.

• Inter-Symbol Interference (ISI): Nyquist criterion for zero ISI: sum(P(f + k/Tb)) = Tb. Nyquist minimum transmission bandwidth = Rb / 2.

• Raised Cosine Filter for band-limited zero ISI: Roll-off factor alpha (0 <= alpha <= 1), actual transmission bandwidth BW = (1 + alpha) * Rb / 2.

• Eye diagram: Interpretation of eye opening (noise margin), slope of eye closure (sensitivity to timing jitter), width of eye (zero-crossing jitter).

SECTION: DIGITAL PASSBAND MODULATION & DETECTION

• Geometric representation of signals: Gram-Schmidt orthogonalization procedure, signal space constellation diagram, basis functions.

• Optimum receiver: Matched filter receiver (impulse response h(t) = s(T - t), maximizes output peak SNR = 2*E / N0), Correlation receiver.

• Binary modulation schemes: Binary Phase Shift Keying (BPSK), Binary Frequency Shift Keying (BFSK, coherent and non-coherent), Binary Amplitude Shift Keying (BASK / OOK).

• M-ary modulation schemes: Quadrature Phase Shift Keying (QPSK, two bits per symbol, constellation points at +-pi/4, +-3pi/4), Quadrature Amplitude Modulation (16-QAM, rectangular constellation, high spectral efficiency), Minimum Shift Keying (MSK, continuous phase FSK, constant envelope, modulation index h = 0.5).

• Bit Error Rate (BER) calculations using Q-function / complementary error function:

• - BPSK: Pe = Q(sqrt(2 * Eb / N0))

• - QPSK: Pe = Q(sqrt(2 * Eb / N0)) (same BER as BPSK, but double spectral efficiency!)

• - Coherent BFSK: Pe = Q(sqrt(Eb / N0)) (3 dB worse power efficiency than BPSK)

• - Non-coherent BFSK: Pe = 0.5 * exp(-Eb / (2*N0))

SECTION: ERROR CONTROL CODING BASICS

• Linear block codes: Generator matrix G of size k x n, Parity check matrix H of size (n-k) x n. Codeword c = m * G, syndrome vector s = r * H^T.

• Minimum Hamming distance (d_min): Detects up to (d_min - 1) errors, corrects up to floor((d_min - 1) / 2) errors.

• Standard Hamming codes: Single-error correcting (7, 4) code with d_min = 3.

• Cyclic codes: Generator polynomial g(x), systematic cyclic encoding using shift registers.

• Convolutional codes: Constraint length K, code rate k/n, generator sequences, state transition diagram, Trellis diagram.

• Maximum Likelihood Viterbi decoding algorithm.

RECOMMENDED STANDARD TEXTBOOKS & REFERENCES

• Communication Systems — Simon Haykin & Michael Moher (Wiley)
• Modern Digital and Analog Communication Systems — B.P. Lathi & Zhi Ding (Oxford)
• Digital Communications — John G. Proakis & Masoud Salehi

HIGH-YIELD GATE EXAM STRATEGY & REVISION TIPS

• QPSK transmits 2 bits per symbol, achieving double the spectral efficiency of BPSK while maintaining the identical bit error probability: Pe = Q(sqrt(2*Eb/N0)).
• Slope overload distortion in Delta Modulation is eliminated when delta >= (2*pi*fm*Am) / fs.
• Matched filter impulse response is a time-reversed and shifted copy of the input signal: h(t) = s(T - t).